

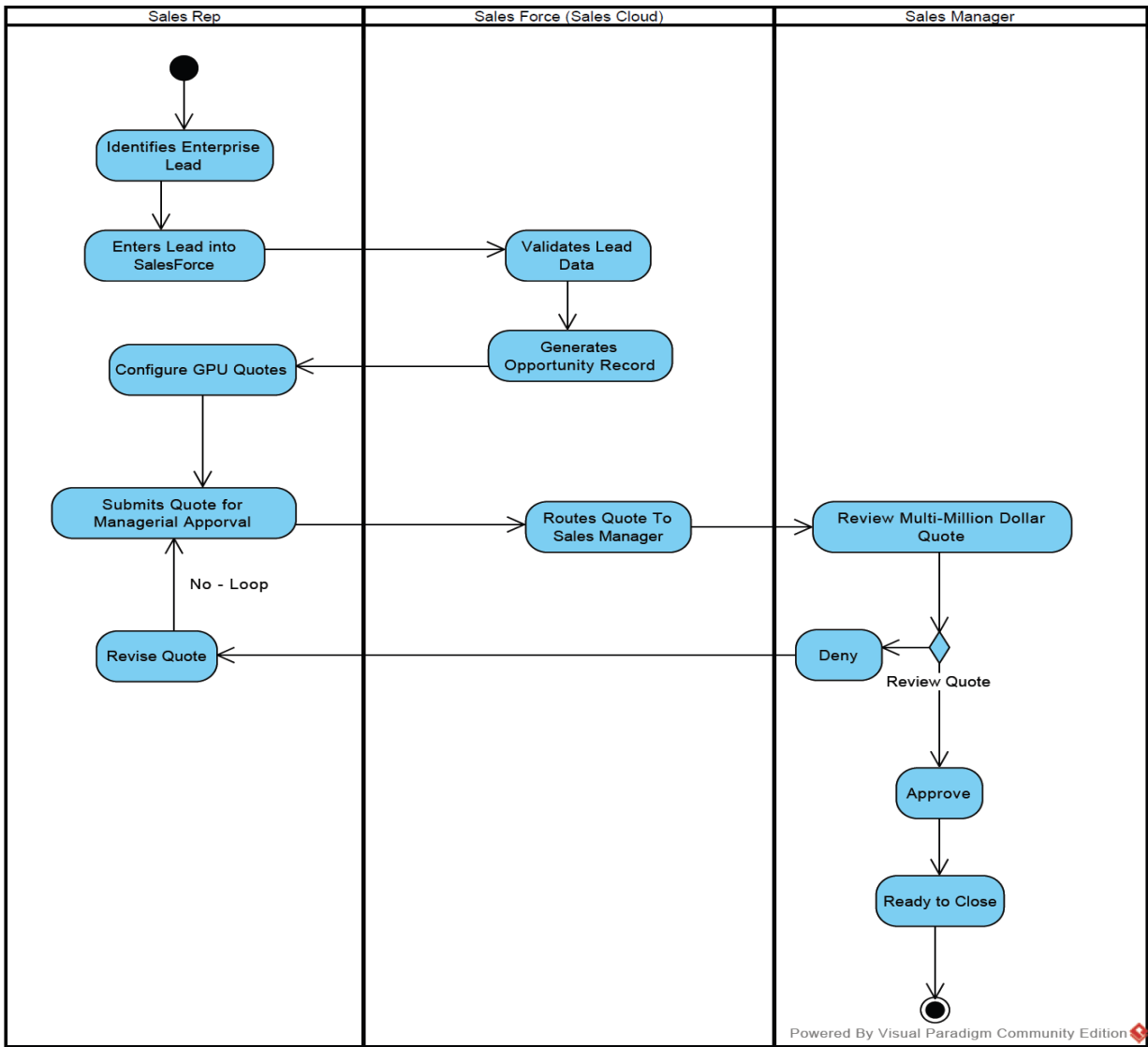
LAB 3: Activity Diagrams and 3 use cases and specification tables

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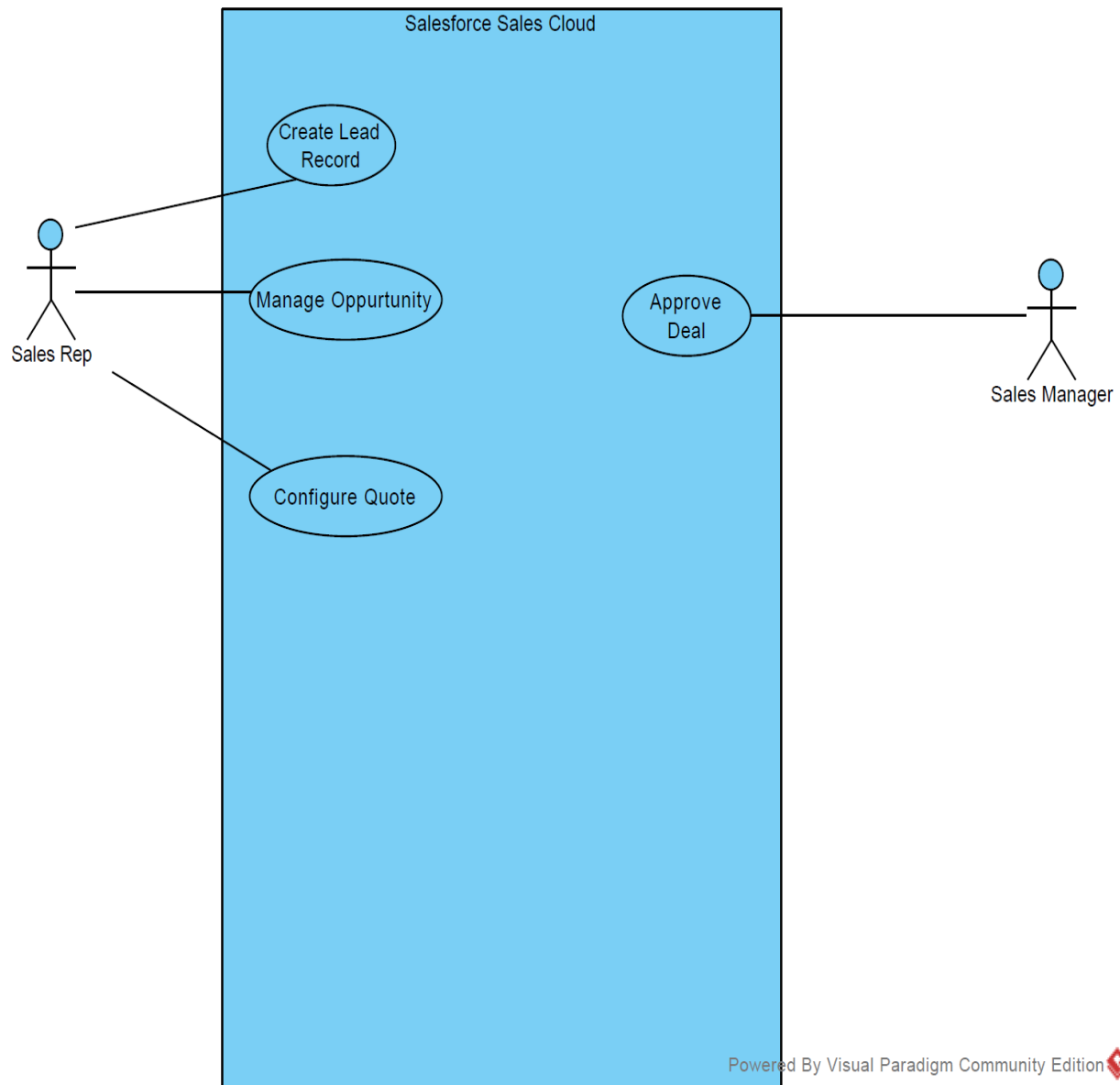
Workflow 1: Complex B2B Sales Cycle (Data Center GPU Clusters)

Description: This workflow maps to NVIDIA's primary revenue driver. It covers the long enterprise sales cycle of selling multimillion-dollar AI infrastructure to tech giants, utilizing Salesforce Sales Cloud to manage the opportunity and approval process.

Process Map 1: B2B Sales Cycle



Use Case 1: Salesforce Sales Cloud



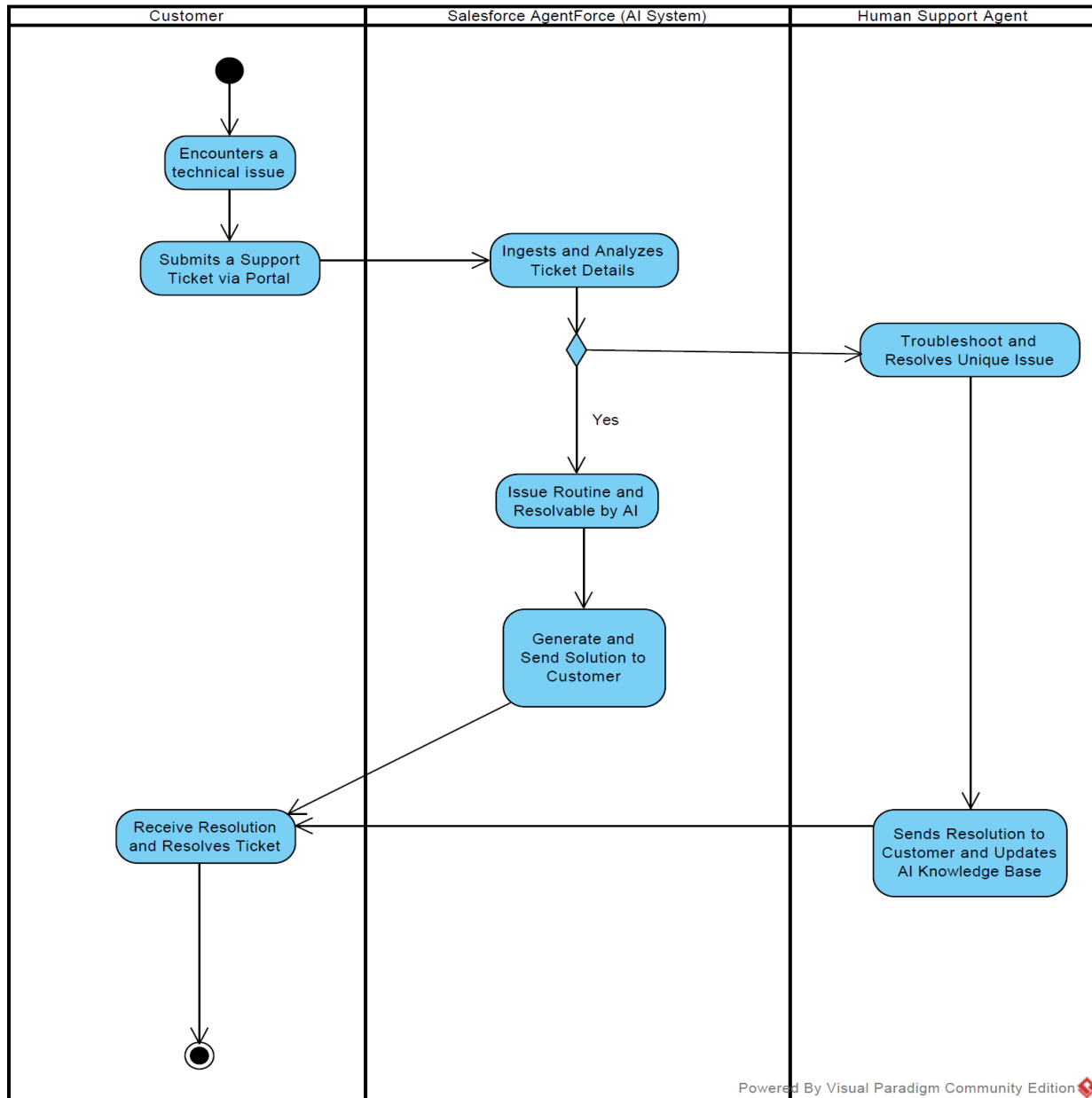
Use Case Specification 1:

Section	Details
Use Case Name	Manage Complex B2B Quote and Approval
Actor(s)	Sales Rep (primary) Salesforce Sales Cloud (system) Sales Manager (secondary)
Summary Description	Allows a Sales Rep to configure a multimillion-dollar GPU cluster quote in Salesforce and route it to a Sales Manager for approval before the deal can be closed.
Priority	Must Have
Status	Medium Level of Details
Pre-Conditions	- Enterprise lead has been entered and validated in Salesforce - An Opportunity record exists in the system - User has authorization to configure quotes
Post-Conditions	- Quote is marked as "Approved" and the Opportunity is "Ready to Close" - OR Quote is rejected and sent back to the rep for revision - System logs the manager's approval decision
Basic Path	- Sales Rep opens the Opportunity in Salesforce - Sales Rep configures the complex GPU cluster quote - Sales Rep submits the quote for managerial approval - System routes the quote to the appropriate regional Sales Manager - Sales Manager reviews the multimillion-dollar quote - Sales Manager selects "Approve" - System updates Opportunity status to "Ready to Close"
Alternative Paths	6a. Sales Manager selects "Reject" → System routes quote back to Sales Rep for revision 2a. Configuration violates pricing rules → System displays error and prevents submission
Business Rules	B1: Quotes over specific dollar thresholds require regional manager approval B2: Validated lead data is required before quote generation
Non-Functional Requirements	NF1: System must route approval requests within 2 seconds NF2: Approval actions must be audited and time-stamped for compliance NF3: Role-based access control is required for managerial views

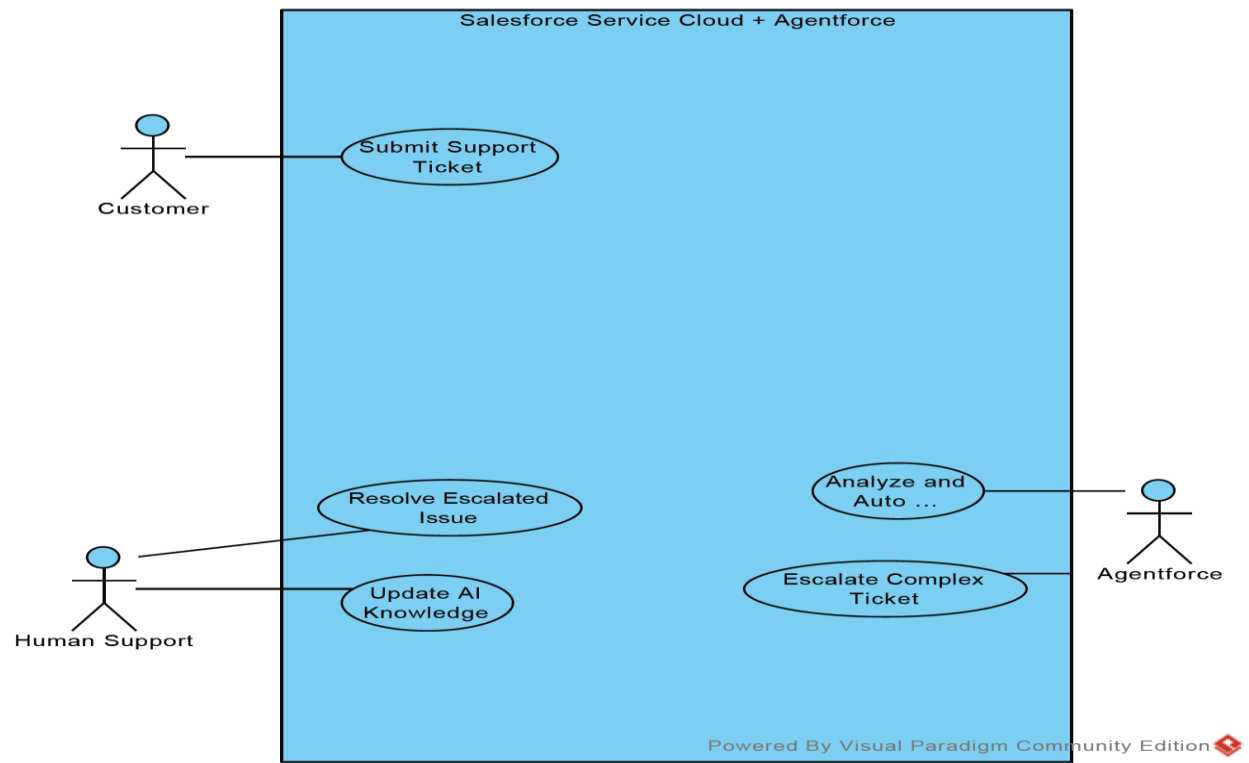
Workflow 2: AI-Automated Customer Support via Agentforce

Description: This workflow leverages the new strategic partnership between Salesforce and NVIDIA, using Agentforce (powered by Nemotron AI) to manage routine developer and enterprise customer service tickets autonomously.

Process Map 2: AI-Automated Customer Support via Agentforce



Use Case Diagram 2:



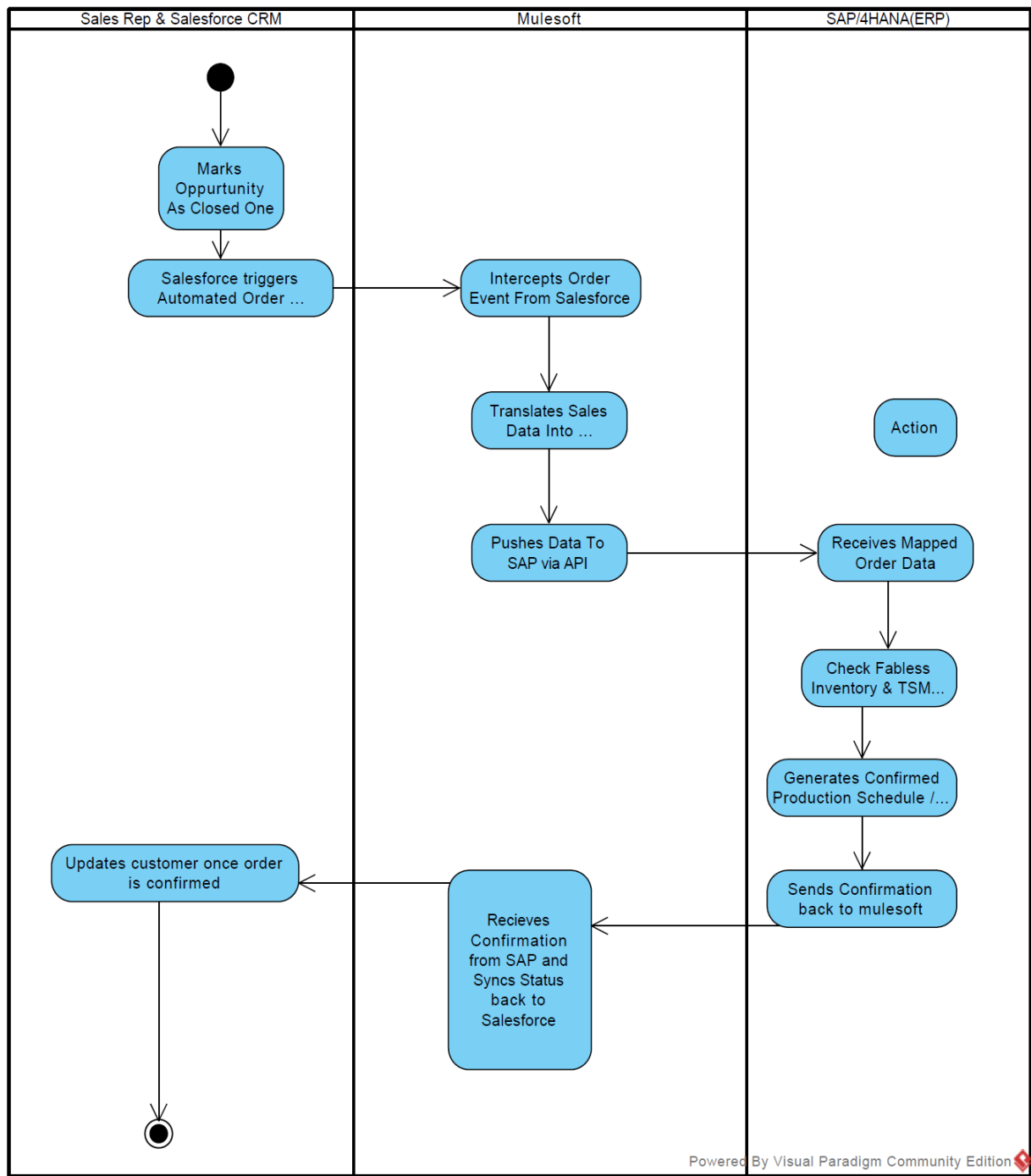
Use Case Specification 2:

Section	Details
Use Case Name	Analyze and Auto-Resolve Support Ticket
Actor(s)	Customer (primary) Salesforce Agentforce (system actor) Human Support Agent (secondary)
Summary Description	Agentforce AI autonomously analyzes incoming customer support tickets to resolve routine issues instantly, escalating only complex problems to human agents.
Priority	Must Have
Status	Medium Level of Details
Pre-Conditions	- Customer is logged into the enterprise portal - Customer has submitted a technical support ticket - AI knowledge base is active and accessible
Post-Conditions	- Ticket is resolved by AI and closed - OR Ticket is escalated and assigned to a Human Support Agent
Basic Path	- Agentforce receives the submitted ticket - Agentforce ingests and analyzes the ticket details/text - Agentforce determines the issue is routine and resolvable by AI - Agentforce generates a solution from its knowledge base - Agentforce sends the solution to the Customer - Customer receives resolution and the ticket is closed
Alternative Paths	3a. Agentforce determines the issue is complex → System escalates and routes ticket to a Human Support Agent → Human Agent troubleshoots, resolves, and updates the AI knowledge base.
Business Rules	B1: AI must meet a specific confidence threshold to auto-resolve an issue B2: Escalated tickets must be routed based on human agent availability and expertise level
Non-Functional Requirements	NF1: AI analysis and initial response must occur within 5 seconds NF2: System must maintain 99.9% uptime for global enterprise clients NF3: The AI knowledge base must be continuously updated in real-time

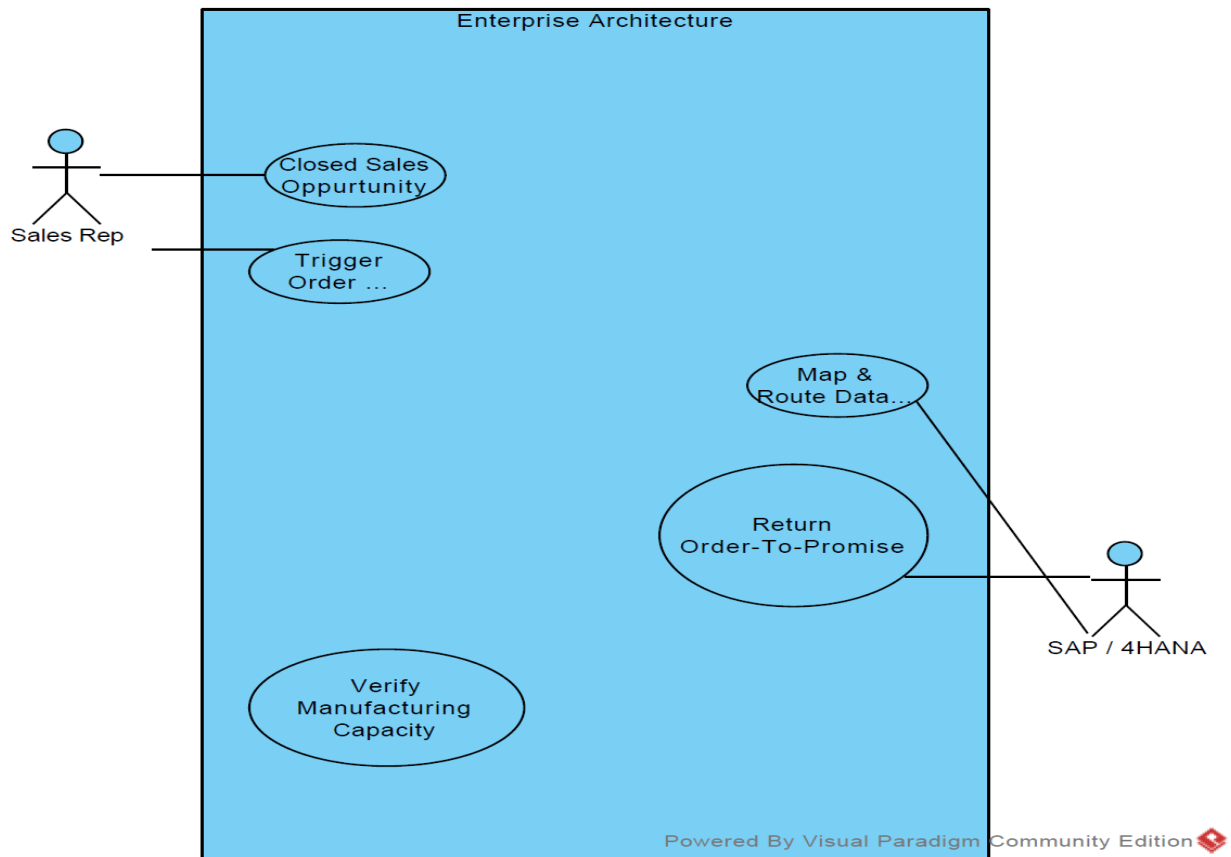
Workflow 3: Fabless Order-to-Promise Integration

Description: This workflow shows how a closed deal in Salesforce flows through MuleSoft into SAP S/4HANA to verify manufacturing capacity at TSMC.

Process Map 3: Fabless Order-to-Promise Integration



Use Case Diagram 3:



Use Case Specification 3:

Section	Details
Use Case Name	Verify Manufacturing Capacity and Confirm Order
Actor(s)	Sales Rep (primary) MuleSoft (middleware/system) SAP S/4HANA (external system actor)
Summary Description	Automates the transfer of a closed-won opportunity from Salesforce through MuleSoft into SAP S/4HANA to verify external manufacturing capacity at TSMC and return a confirmed delivery date.
Priority	Must Have
Status	Medium Level of Details
Pre-Conditions	- Opportunity is marked as "Closed Won" in Salesforce - MuleSoft API connections to SAP S/4HANA are active and authenticated - Data mapping templates are configured
Post-Conditions	- Confirmed production schedule is returned to Salesforce - Customer is updated with an accurate delivery date - Order is officially logged in SAP
Basic Path	- Sales Rep marks Opportunity as "Closed Won" in Salesforce - Salesforce triggers an automated "Order Creation" event - MuleSoft intercepts the event and translates data into SAP format - MuleSoft pushes the data payload to SAP S/4HANA via API - SAP S/4HANA checks fabless inventory and TSMC production capacity - SAP S/4HANA generates a confirmed Order-to-Promise delivery date - SAP sends confirmation back to MuleSoft - MuleSoft syncs the status back to Salesforce - Salesforce updates the Customer with the confirmed order
Alternative Paths	5a. Insufficient TSMC capacity → SAP returns a delayed delivery date → Salesforce alerts Sales Rep to renegotiate timeline 4a. API connection times out → MuleSoft queues the payload and retries automatically
Business Rules	B1: Orders cannot be confirmed to the customer until TSMC capacity is verified in SAP B2: Data payloads must strictly conform to SAP data mapping standards
Non-Functional Requirements	NF1: End-to-end integration response time between the 3 systems must be under 10 seconds NF2: All API calls must be encrypted using enterprise-grade security NF3: Middleware must log all data transfer events for troubleshooting